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MyAgent.py – Complete Tool Reference
Operating System & Provider Differences
(Windows / macOS) x (Anthropic / OpenAI / Gemini)
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1. OVERVIEW & TOOL ASSEMBLY
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MyAgent.py defines tools in four global lists plus one dynamic tool. The `_get_tools()` method assembles the final tool set sent to the API based on UI toggle states and the active provider.

|               |           |                                                                                    |
|---------------|-----------|------------------------------------------------------------------------------------|
| TOOLS         | 5 tools.  | Always included. Two are replaced for Anthropic/OpenAI by server-side equivalents. |
| DESKTOP_TOOLS | 13 tools. | Included when Desktop checkbox is ON.                                              |
| BROWSER_TOOLS | 11 tools. | Included when Browser checkbox is ON.                                              |
| META_TOOLS    | 3 tools.  | Included when Meta checkbox is ON.                                                 |
| get_skill     | 1 tool.   | Added dynamically when any skill has mode="on_demand".                             |

### ASSEMBLY LOGIC (`_get_tools()`):

1. Start with a deep copy of TOOLS (5 tools).
2. IF provider is Anthropic or OpenAI:  
    REMOVE local `web_search` and `fetch_webpage`.  
    (These providers supply server-side equivalents with better quality.)  
IF provider is Gemini:  
    KEEP all 5 tools.  
    (Gemini cannot combine built-in tools with custom function declarations -- an API restriction.)
3. IF Desktop checkbox is ON and `pyautogui` is installed:

- Add 13 DESKTOP\_TOOLS (screenshot description is dynamically patched with actual display resolution info).
- 4. IF Browser checkbox is ON:
  - Add 11 BROWSER\_TOOLS.
- 5. IF Meta checkbox is ON:
  - Add 3 META\_TOOLS.
- 6. IF any skill has mode="on\_demand":
  - Add get\_skill tool with enum of those skill names.

#### PARALLEL EXECUTION:

Tools in PARALLEL\_SAFE = {web\_search, fetch\_webpage, csv\_search, get\_skill} run concurrently via ThreadPoolExecutor when multiple are requested in one turn. All other tools (desktop, browser, meta, run\_command, user\_prompt) run sequentially.

## 2. CORE TOOLS (TOOLS)

These 5 tools are always included in the base tool list. Two (web\_search, fetch\_webpage) are removed for Anthropic and OpenAI providers because those providers supply server-side replacements.

### 2.1 web\_search

#### Description:

Search the web for information. Use this to find current information, answer questions about recent events, look up facts, or find relevant websites. Always prefer searching before guessing.

#### Parameters:

query (string, REQUIRED) The search query.

#### PROVIDER DIFFERENCES:

|           |                                                                                                                                                          |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anthropic | EXCLUDED from local tools. Replaced by server-side web_search_20250305 (see Section 7). The model's own web search runs server-side with richer results. |
| OpenAI    | EXCLUDED from local tools. Replaced by server-side web_search_preview (see Section 7).                                                                   |
| Gemini    | INCLUDED. Runs LOCALLY via DuckDuckGo (DDGS library). Calls DDGS().text(query, max_results=5), returning title, URL, and snippet for each result.        |

OS DIFFERENCES: None. The local implementation (DuckDuckGo) is identical on both Windows and macOS.

### 2.2 fetch\_webpage

---

Description:

Fetch the full content of a specific webpage URL. Use this after `web_search` to read a page in detail, or when the user provides a specific URL.

Parameters:

`url` (string, REQUIRED) The URL to fetch.

Implementation:

Uses `httpx.get()` with `follow_redirects=True`, `timeout=15s`. HTML content is parsed to plain text via `HTMLTextExtractor`. Output truncated at 20,000 characters.

PROVIDER DIFFERENCES:

|           |                                                        |
|-----------|--------------------------------------------------------|
| Anthropic | EXCLUDED (server-side web search handles URL content). |
| OpenAI    | EXCLUDED (server-side web search handles URL content). |
| Gemini    | INCLUDED. Runs locally as described above.             |

OS DIFFERENCES: None.

---

## 2.3 `run_command`

---

Description:

Execute a command on the local machine and return its output. Use this for system tasks like listing files, checking processes, reading/writing files, getting system info, running scripts, installing software, or any other local operation. Commands run with the current user's permissions. On Windows this runs PowerShell; on macOS this runs bash. IMPORTANT: When launching GUI applications, use `Start-Process` (Windows) or `'open -a'` (macOS) so the command returns immediately instead of blocking.

Parameters:

`command` (string, REQUIRED) The command to execute.

PROVIDER DIFFERENCES: None. Identical across all three providers.

OS DIFFERENCES:

|         |                                                                                                                                                                                               |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Windows | Shell: ["powershell", "-NoProfile", "-Command", command]<br>Subprocess flag: <code>CREATE_NO_WINDOW</code> to hide console.<br>Timeout: 30 seconds.<br>Output truncated at 20,000 characters. |
| macOS   | Shell: ["/bin/bash", "-c", command]<br>No special subprocess flags.<br>Timeout: 30 seconds.<br>Output truncated at 20,000 characters.                                                         |

Before execution, every command is checked against the safety system (see Section 11 for OS-specific blocked and confirmation patterns).

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## 2.4 csv\_search

---

### Description:

Search a delimited text file (CSV, TSV, TXT, etc.) for records matching a value. The file must have a header row. You can search a specific column or all columns. Returns matching rows as formatted text.

### Parameters:

|              |                     |                                                                                                                                        |
|--------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| file_path    | (string, REQUIRED)  | Absolute path to the file.                                                                                                             |
| search_value | (string, REQUIRED)  | The value to search for.                                                                                                               |
| delimiter    | (string, optional)  | Column delimiter. ',' for CSV (default), '\t' for TSV, ' ' for pipe, ';' for semicolons. Auto-detects from file content if omitted.    |
| column       | (string, optional)  | Column heading to search in. Searches all columns if omitted.                                                                          |
| match_mode   | (string, optional)  | How to match: 'contains' (default), 'exact', or 'starts_with'. All modes are case-insensitive.<br>Enum: [contains, exact, starts_with] |
| max_results  | (integer, optional) | Max matching rows (default 50).                                                                                                        |

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None.

---

## 2.5 user\_prompt

---

### Description:

Pause execution and display a message to the user, then wait for their response. You MUST use this tool whenever you need the user to do something or when you need information only the user can provide. NEVER just output text asking the user something -- that ends your turn and they cannot reply. This tool is the ONLY way to communicate with the user and receive a response.

### Parameters:

|         |                    |                                     |
|---------|--------------------|-------------------------------------|
| message | (string, REQUIRED) | The message or question to display. |
|---------|--------------------|-------------------------------------|

PROVIDER DIFFERENCES: None.

### OS DIFFERENCES:

|         |                                                                                                                                                  |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Windows | Dialog uses .transient(parent) for proper parent-child window relationship.                                                                      |
| macOS   | Dialog SKIPS .transient(parent). macOS window manager restricts transient windows to the parent's screen, preventing drag to secondary monitors. |

+-----+

Headless mode note: With --headless flag, the dialog skips both transient() and grab\_set() so it floats as a standalone window on both platforms.

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### 3. DESKTOP TOOLS (DESKTOP\_TOOLS) -- 13 tools

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Available when the Desktop checkbox is ON and pyautogui/dependencies are installed (\_HAS\_DESKTOP is True). These tools use pyautogui for cross-platform desktop automation. pyautogui.PAUSE is set to 0.1 seconds globally.

#### COORDINATE SYSTEM:

All coordinate-based tools (mouse\_click, mouse\_scroll, mouse\_drag, read\_screen\_text, find\_image\_on\_screen) use "image coordinates" -- pixel positions as they appear in the most recent screenshot image. These are automatically mapped to screen coordinates using the formula:

```
screen_x = round(image_x * _screenshot_scale) + _screenshot_offset[0]
screen_y = round(image_y * _screenshot_scale) + _screenshot_offset[1]
```

Where:

```
_screenshot_scale = resize ratio applied when fitting to API limits
_screenshot_offset = (x, y) origin of the captured display
_screenshot_dims = (width, height) of the last sent image
```

-----

#### 3.1 screenshot

-----

##### Description:

(Dynamically generated at runtime with actual display info.)  
Captures the desktop as a PNG image. Supports multi-monitor setups and region capture. The runtime description lists available displays and their resolutions, instructing the model to:

- (1) Take screenshot with no display param to see all screens
- (2) Call screenshot with the specific display number
- (3) Use those coordinates for subsequent clicks

##### Parameters:

|         |                     |                                                                                                                                     |
|---------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| display | (integer, optional) | Which display to capture.<br>0 = primary, 1 = secondary, etc.<br>Omit to capture ALL displays as separate images (one per display). |
| x       | (integer, optional) | Left edge of region to capture.                                                                                                     |
| y       | (integer, optional) | Top edge of region to capture.                                                                                                      |
| width   | (integer, optional) | Width of region to capture.                                                                                                         |
| height  | (integer, optional) | Height of region to capture.                                                                                                        |

##### PROVIDER DIFFERENCES (image resize limits):

+-----+

|           |                                                         |  |
|-----------|---------------------------------------------------------|--|
| Anthropic | Max 1568px long edge, 1,150,000 total pixels (1.15 MP). |  |
|-----------|---------------------------------------------------------|--|

+-----+

|        |                                                                                                                                                                                                                                                                                                     |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OpenAI | Max 2048px long edge, 2,000,000 total pixels (2.0 MP).                                                                                                                                                                                                                                              |
| Gemini | Max 2048px long edge, 2,000,000 total pixels (2.0 MP).<br>Additionally, coordinate hints are appended to the screenshot tool description via <code>_GEMINI_COORD_HINTS</code> to help Gemini understand the coordinate system (Gemini is less reliable with coordinates without explicit guidance). |

## OS DIFFERENCES:

### DISPLAY ENUMERATION:

|         |                                                                                                                                         |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Windows | <code>EnumDisplayMonitors</code> via <code>ctypes.windll.user32</code> .<br>Returns monitor rectangles sorted with primary (0,0) first. |
| macOS   | <code>CGGetActiveDisplayList</code> via <code>CoreGraphics</code> ctypes (up to 16 displays). Display 0 = primary at origin (0,0).      |

### FULL DISPLAY CAPTURE:

|         |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Windows | <code>ImageGrab.grab(bbox=(l,t,r,b), all_screens=True)</code> for each display's bounding rectangle.                                                                                                                                                                                                                                                                                                                        |
| macOS   | <code>pyautogui.screenshot()</code> only captures the primary display. Uses Quartz <code>CGWindowListCreateImage()</code> with per-display <code>CGRect</code> for individual display capture. Flags: <code>kCGWindowListOptionOnScreenOnly</code> , <code>kCGNullWindowID</code> , <code>kCGWindowImageDefault</code> . Raw BGRA pixel data is extracted via <code>CGDataProviderCopyData</code> and converted to PIL RGB. |

### REGION CAPTURE:

|         |                                                                                                                                                                                                                            |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Windows | <code>ImageGrab.grab(bbox=(x,y,x+w,y+h), all_screens=True)</code> .<br>Works across multi-monitor setups including negative coords.                                                                                        |
| macOS   | <code>pyautogui.screenshot(region=(x,y,w,h))</code> .<br>Retina DPI correction: if physical pixel width differs from requested logical width, the image is resized to logical dimensions to maintain coordinate alignment. |

### DPI / RETINA HANDLING:

|         |                                                                                                                                                                                                                                                             |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Windows | <code>SetProcessDpiAwareness(2)</code> called at startup. With DPI awareness enabled, all coordinates are physical pixels -- no additional DPI alignment needed.                                                                                            |
| macOS   | Quartz captures at physical pixel resolution (2x on Retina). <code>_capture_single_display()</code> detects this by comparing physical image dimensions to logical dimensions from <code>_get_display_rects()</code> . If they differ, the image is resized |

|  |                                                         |
|--|---------------------------------------------------------|
|  | to logical size so pyautogui's coordinate space aligns. |
|--|---------------------------------------------------------|

### 3.2 mouse\_click

#### Description:

Click the mouse at the given (x, y) position. Take a screenshot first to identify the correct coordinates -- use pixel positions as seen in the screenshot image. Coordinates are automatically mapped to the actual screen. Supports left/right/middle button and single/double click.

#### Parameters:

|        |                     |                                                             |
|--------|---------------------|-------------------------------------------------------------|
| x      | (integer, REQUIRED) | X coordinate to click.                                      |
| y      | (integer, REQUIRED) | Y coordinate to click.                                      |
| button | (string, optional)  | Mouse button. Enum: [left, right, middle]<br>Default: left. |
| clicks | (integer, optional) | Number of clicks. Enum: [1, 2]<br>Default: 1.               |

#### PROVIDER DIFFERENCES:

|           |                                                                                                                                                                                                                           |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anthropic | Schema used as-is. Integer enums preserved.                                                                                                                                                                               |
| OpenAI    | Schema used as-is. Integer enums preserved.                                                                                                                                                                               |
| Gemini    | Integer enums for 'clicks' are STRIPPED (Gemini only supports string enums). The parameter reverts to plain integer type. Coordinate hints are appended to the description reminding the model to use image pixel coords. |

OS DIFFERENCES: None. pyautogui.click() is cross-platform. The coordinate mapping (scale + offset) is identical on both platforms -- only the screenshot capture differs (handled in screenshot tool).

Implementation detail: Coordinates are clamped to screenshot bounds (\_screenshot\_dims) with a warning if out of range.

### 3.3 type\_text

#### Description:

Type text at the current cursor position. Click on an input field first to focus it, then use this tool to type. Uses clipboard paste for non-ASCII characters.

#### Parameters:

|          |                    |                                             |
|----------|--------------------|---------------------------------------------|
| text     | (string, REQUIRED) | The text to type.                           |
| interval | (number, optional) | Seconds between keystrokes (default: 0.02). |

PROVIDER DIFFERENCES: None.

#### OS DIFFERENCES:

|         |                                                                                                                         |
|---------|-------------------------------------------------------------------------------------------------------------------------|
| Windows | ASCII text: <code>pyautogui.write()</code> (cross-platform).<br>Non-ASCII: clipboard paste via <code>Ctrl+V</code> .    |
| macOS   | ASCII text: <code>pyautogui.write()</code> (cross-platform).<br>Non-ASCII: clipboard paste via <code>Command+V</code> . |

Both platforms use `pyperclip` for clipboard access when pasting non-ASCII.

### 3.4 `press_key`

#### Description:

Press a key or key combination. Use '+' to combine keys. Examples: 'enter', 'tab', 'escape', 'ctrl+c', 'ctrl+shift+s', 'alt+tab', 'command+c', 'command+q'. Key names follow `pyautogui` naming. Use `ctrl/alt/win` on Windows, `command/option` on macOS.

#### Parameters:

`keys` (string, REQUIRED) Key or combo, e.g. 'enter', 'ctrl+c'.

PROVIDER DIFFERENCES: None.

#### OS DIFFERENCES (key name normalisation):

|         |                                                                                                                                                                                                                                                                                                                                  |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Windows | <code>windows</code> -> <code>win</code><br><code>control</code> -> <code>ctrl</code><br><code>return</code> -> <code>enter</code><br><code>esc</code> -> <code>escape</code>                                                                                                                                                    |
| macOS   | <code>windows</code> -> <code>command</code><br><code>win</code> -> <code>command</code><br><code>cmd</code> -> <code>command</code><br><code>option</code> -> <code>alt</code><br><code>control</code> -> <code>ctrl</code> (unchanged)<br><code>return</code> -> <code>enter</code><br><code>esc</code> -> <code>escape</code> |

The '+' separator is split, each part is normalised via the above aliases, then single keys use `pyautogui.press()` and combinations use `pyautogui.hotkey()`.

### 3.5 `mouse_scroll`

#### Description:

Scroll the mouse wheel. Positive clicks = scroll up, negative = scroll down. Optionally specify (x, y) to scroll at a specific position.

#### Parameters:



|        |                     |                                            |
|--------|---------------------|--------------------------------------------|
| clicks | (integer, REQUIRED) | Scroll amount: positive=up, negative=down. |
| x      | (integer, optional) | X coordinate to scroll at.                 |
| y      | (integer, optional) | Y coordinate to scroll at.                 |

PROVIDER DIFFERENCES: None. (Gemini strips integer enums but this tool has no enum parameters.)

OS DIFFERENCES: None. pyautogui.scroll() is cross-platform. Coordinate mapping uses same scale/offset as mouse\_click.

---

### 3.6 open\_application

---

#### Description:

Open an application by common name or full path. Known names: chrome, firefox, edge, safari, notepad, notepad++, calculator, terminal, finder, excel, word, vscode, spotify, discord, slack, teams. Or provide a full executable path. Use 'args' to pass arguments.

#### Parameters:

|      |                    |                                       |
|------|--------------------|---------------------------------------|
| name | (string, REQUIRED) | App name or full path to executable.  |
| args | (string, optional) | Arguments to pass (e.g. a file path). |

PROVIDER DIFFERENCES: None.

OS DIFFERENCES (KNOWN\_APPS dictionary):

#### Windows mappings:

|            |                     |
|------------|---------------------|
| chrome     | -> start chrome     |
| firefox    | -> start firefox    |
| edge       | -> start msedge     |
| notepad    | -> notepad          |
| notepad++  | -> start notepad++  |
| calculator | -> calc             |
| calc       | -> calc             |
| excel      | -> start excel      |
| word       | -> start winword    |
| powerpoint | -> start powerpnt   |
| explorer   | -> explorer         |
| cmd        | -> start cmd        |
| powershell | -> start powershell |
| vscode     | -> code             |
| code       | -> code             |
| spotify    | -> start spotify:   |
| discord    | -> start discord:   |
| slack      | -> start slack:     |
| teams      | -> start msteams:   |

#### macOS mappings:

|            |                             |
|------------|-----------------------------|
| chrome     | -> open -a 'Google Chrome'  |
| firefox    | -> open -a Firefox          |
| edge       | -> open -a 'Microsoft Edge' |
| safari     | -> open -a Safari           |
| calculator | -> open -a Calculator       |
| calc       | -> open -a Calculator       |
| terminal   | -> open -a Terminal         |

```
finder -> open .
explorer -> open .
vscode -> code
code -> code
spotify -> open -a Spotify
discord -> open -a Discord
slack -> open -a Slack
teams -> open -a 'Microsoft Teams'
```

Note: Windows has notepad, notepad++, powerpoint, cmd, powershell that macOS lacks. macOS has safari and terminal that Windows lacks. Some entries like vscode/code and explorer/finder overlap with platform-appropriate commands.

If name is not in KNOWN\_APPS, it is treated as a direct command.

---

### 3.7 find\_window

---

#### Description:

Find windows matching a title pattern. Returns window titles, positions, and sizes. Optionally activate (bring to foreground) the first match.

#### Parameters:

```
title (string, REQUIRED) Window title or partial text to search.
activate (boolean, optional) Bring first match to foreground
 (default: false).
```

PROVIDER DIFFERENCES: None.

#### OS DIFFERENCES:

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Windows | ENUMERATION: <code>pygetwindow.getWindowsWithTitle(title)</code> .<br>Returns windows with title, left, top, width, height.<br>ACTIVATION: <code>win.restore()</code> if minimised, then <code>win.activate()</code> .                                                                                                                                                                                                                                  |
| macOS   | ENUMERATION: Quartz <code>CGWindowListCopyWindowInfo()</code> with flags<br><code>kCGWindowListOptionOnScreenOnly</code><br><code>kCGWindowListExcludeDesktopElements</code> .<br>Extracts <code>kCGWindowOwnerName</code> (app), <code>kCGWindowName</code> (title),<br><code>kCGWindowBounds</code> (position/size).<br>Full title: "{app} -- {name}" if name else app.<br>ACTIVATION: osascript AppleScript:<br>tell application "{app}" to activate |

Matching is case-insensitive substring on both platforms.

---

### 3.8 clipboard\_read

---

#### Description:

Read the current text contents of the clipboard.

Parameters: None.

Implementation: `tkinter root.clipboard_get()` (cross-platform).  
Returns error if clipboard is empty or contains non-text data.

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None.

---

### 3.9 clipboard\_write

---

Description:

Write text to the clipboard, replacing any current content.

Parameters:

`text` (string, REQUIRED) The text to place on the clipboard.

Implementation: `root.clipboard_clear()` then `root.clipboard_append(text)`.  
Returns preview (first 100 chars) and total character count.

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None.

---

### 3.10 wait\_for\_window

---

Description:

Wait until a window with the given title appears, polling every 0.5 seconds. Returns the window info once found, or times out.

Parameters:

`title` (string, REQUIRED) Window title or partial text to wait for.  
`timeout` (number, optional) Maximum seconds to wait (default: 10).

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: Uses the same platform-specific `_find_windows_by_title()` as `find_window` (pygetwindow on Windows, CGWindowListCopyWindowInfo on macOS). See `find_window` above.

---

### 3.11 read\_screen\_text (OCR)

---

Description:

Read text from a region of the screen using OCR. Specify the region as `x`, `y`, `width`, `height` using coordinates from the screenshot image.

Parameters:

`x` (integer, REQUIRED) Left edge of region.  
`y` (integer, REQUIRED) Top edge of region.  
`width` (integer, REQUIRED) Width of region.

height (integer, REQUIRED) Height of region.

PROVIDER DIFFERENCES: None.

OS DIFFERENCES:

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Windows | CAPTURE: ImageGrab.grab(bbox=..., all_screens=True).<br>OCR ENGINE: winocr library (Windows built-in OCR).<br>asyncio.run(winocr.recognize_pil(img, lang="en"))<br>Returns result.text.strip().                                                                                                                                                                                                                                        |
| macOS   | CAPTURE: pyautogui.screenshot(region=...).<br>OCR ENGINE: Apple Vision framework via objc bridge.<br>1. Load Vision bundle: objc.loadBundle("Vision", ...)<br>2. Convert PIL image to PNG bytes -> NSData -> UIImage<br>3. Create VNImageRequestHandler with UIImage<br>4. Run VNRecognizeTextRequest (recognition level 0 =<br>accurate mode)<br>5. Extract candidate strings from observations<br>Returns newline-joined text lines. |

Coordinate transformation uses same scale/offset mapping as screenshot region capture.

---

### 3.12 find\_image\_on\_screen

---

Description:

Find an image on the screen by matching a reference image file. Returns the center coordinates if found. Useful for finding buttons or icons.

Parameters:

image\_path (string, REQUIRED) Absolute path to reference image (PNG/JPG).  
confidence (number, optional) Match threshold 0.0-1.0 (default: 0.8).

Implementation:

pyautogui.locateOnScreen(confidence=...). Returns center point in both screen coordinates and image coordinates:

```
img_cx = round((screen_cx - offset_x) / scale)
img_cy = round((screen_cy - offset_y) / scale)
```

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None. pyautogui.locateOnScreen() is cross-platform.

---

### 3.13 mouse\_drag

---

Description:

Drag the mouse from one point to another. Useful for drag-and-drop, resizing windows, moving sliders, drawing, etc.

Parameters:

start\_x (integer, REQUIRED) Starting X coordinate.

|          |                     |                                                    |
|----------|---------------------|----------------------------------------------------|
| start_y  | (integer, REQUIRED) | Starting Y coordinate.                             |
| end_x    | (integer, REQUIRED) | Ending X coordinate.                               |
| end_y    | (integer, REQUIRED) | Ending Y coordinate.                               |
| duration | (number, optional)  | Duration of drag in seconds (default: 0.5).        |
| button   | (string, optional)  | Mouse button: left, right, middle (default: left). |

#### Implementation:

Transforms both start and end coordinates via scale/offset, then:  
 pyautogui.moveTo(start, 0.1s) -> mouseDown() -> moveTo(end, duration)  
 -> mouseUp()

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None. pyautogui drag methods are cross-platform.

## 4. BROWSER TOOLS (BROWSER\_TOOLS) -- 11 tools

Available when the Browser checkbox is ON. Uses Playwright to connect to Google Chrome or Microsoft Edge via Chrome DevTools Protocol (CDP) on port 9222. The browser is launched automatically if not already running.

### BROWSER DISCOVERY & CONNECTION:

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Windows | Searches for browser executables at:<br>%ProgramFiles%\Google\Chrome\Application\chrome.exe<br>%ProgramFiles(x86)%\Google\Chrome\Application\chrome.exe<br>%LocalAppData%\Google\Chrome\Application\chrome.exe<br>%ProgramFiles(x86)%\Microsoft\Edge\Application\msedge.exe<br>%ProgramFiles%\Microsoft\Edge\Application\msedge.exe<br>%LocalAppData%\Microsoft\Edge\Application\msedge.exe<br>Uses subprocess.Popen with CREATE_NO_WINDOW flag. |
| macOS   | Searches for browser executables at:<br>/Applications/Google Chrome.app/Contents/MacOS/...<br>/Applications/Microsoft Edge.app/Contents/MacOS/...<br>Uses subprocess.Popen (no special flags).                                                                                                                                                                                                                                                   |

Both platforms: Launches with --remote-debugging-port=9222 --no-first-run and a temp user-data-dir. Waits up to 15 seconds (30 polls x 0.5s) for the CDP port to become available. Connects via Playwright's chromium.connect\_over\_cdp("http://127.0.0.1:9222").

### 4.1 browser\_open

#### Description:

Open or connect to Google Chrome or Microsoft Edge and navigate to a URL. Uses the user's real browser profile with all cookies, logins, and extensions. If the browser isn't running, it will be launched automatically. Call this first before using any other browser tools.

Parameters:

url (string, REQUIRED) The URL to navigate to.

PROVIDER DIFFERENCES: None.

OS DIFFERENCES:

|         |                                                                                                |
|---------|------------------------------------------------------------------------------------------------|
| Windows | Kills existing browser: PowerShell<br>taskkill /F /IM msedge.exe 2>\$null; Start-Sleep -Ms 500 |
| macOS   | Kills existing browser:<br>pkill -f "Microsoft Edge" + time.sleep(0.5)                         |

After cleanup, calls `_ensure_browser()` then `page.goto(url, wait_until="domcontentloaded", timeout=30000)`.

## 4.2 browser\_navigate

Description:

Navigate the current browser page to a new URL.

Parameters:

url (string, REQUIRED) The URL to navigate to.

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None.

## 4.3 browser\_click

Description:

Click an element on the page. Use a CSS selector or provide visible text to find and click the element. Prefer selectors when possible.

Parameters:

selector (string, optional) CSS selector of the element to click.  
text (string, optional) Visible text of the element to click  
(used if selector is not provided).

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None.

## 4.4 browser\_fill

Description:

Fill a form field with text. Clears any existing value and types instantly (not character-by-character).

Parameters:

selector (string, REQUIRED) CSS selector of the input field.  
value (string, REQUIRED) The text to fill.

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None.

---

#### 4.5 browser\_get\_text

---

Description:

Get the text content of the page or a specific element. If no selector is given, returns full page text. Output truncated at 20,000 characters.

Parameters:

selector (string, optional) CSS selector. Omit for full page text.

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None.

---

#### 4.6 browser\_run\_js

---

Description:

Execute JavaScript code on the current page and return the result. The code runs in the page context. Use 'return' to get a value back.

Parameters:

code (string, REQUIRED) JavaScript to execute.

Implementation: Wraps code as arrow function if needed, runs via page.evaluate(), returns JSON-serialised result.

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None.

---

#### 4.7 browser\_screenshot

---

Description:

Take a screenshot of the current browser page. Returns an image.

Parameters: None.

Implementation: page.screenshot(type="png"), resized to max 2048px width if needed. Returns base64-encoded PNG.

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None.

---

#### 4.8 browser\_close

---

Description:

Disconnect from the browser. The browser stays open -- only the automation connection is closed.

Parameters: None.

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None.

---

#### 4.9 browser\_wait\_for

---

Description:

Wait for an element matching a CSS selector to appear on the page. Returns the element's text content once found, or times out.

Parameters:

selector (string, REQUIRED) CSS selector to wait for.  
timeout (integer, optional) Max milliseconds to wait (default: 10000).

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None.

---

#### 4.10 browser\_select

---

Description:

Select an option from a <select> dropdown element. Specify by value attribute or visible label text.

Parameters:

selector (string, REQUIRED) CSS selector for the <select> element.  
value (string, optional) Option value attribute.  
label (string, optional) Visible text of the option.

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None.

---

#### 4.11 browser\_get\_elements

---

Description:

Get information about elements matching a CSS selector. Returns tag name, text content, key attributes, and visibility for each match.

Parameters:

selector (string, REQUIRED) CSS selector to query.  
limit (integer, optional) Max elements to return (default: 10).

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None.



## =====

### 5. META TOOLS (META\_TOOLS) -- 3 tools

## =====

Available when the Meta checkbox is ON. These tools allow the agent to manage its own instruction library, skills, and spawn child agent processes.

#### -----

#### 5.1 manage\_instructions

#### -----

##### Description:

Manage the saved agent instruction library on disk. You CAN read and update the currently-running instruction -- changes are saved to disk and take effect the next time it is loaded (the live session is not affected).

##### Parameters:

|                  |                     |                                                                         |
|------------------|---------------------|-------------------------------------------------------------------------|
| action           | (string, REQUIRED)  | Enum: [list, read, create, update, delete]                              |
| name             | (string, optional)  | Instruction name (required for all except list).                        |
| text             | (string, optional)  | Instruction content (required for create).                              |
| desktop          | (boolean, optional) | Enable desktop tools.                                                   |
| browser          | (boolean, optional) | Enable browser tools.                                                   |
| meta             | (boolean, optional) | Enable meta tools.                                                      |
| provider         | (string, optional)  | Enum: [Anthropic, OpenAI, Gemini]                                       |
| model            | (string, optional)  | Model name.                                                             |
| temperature      | (number, optional)  | Temperature 0.0-1.0.                                                    |
| thinking_enabled | (boolean, optional) | Enable thinking/reasoning.                                              |
| thinking_effort  | (string, optional)  | Enum: [low, medium, high, max] 'max' only for Anthropic Opus.           |
| thinking_budget  | (integer, optional) | Thinking token budget.                                                  |
| thinking_mode    | (string, optional)  | Enum: [off, adaptive_low, adaptive_medium, adaptive_high, adaptive_max] |
| skill_modes      | (object, optional)  | {skill_name: mode_string} map.                                          |

##### Notes:

- "create" inherits current provider/model/thinking settings.
- "read" returns all model params.
- "update" only changes listed fields; omitted fields keep their values.

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None.

#### -----

#### 5.2 manage\_skills

#### -----

##### Description:

Manage the shared skills library on disk. Skills can be injected into system prompts (enabled), retrieved on demand (on\_demand), or inactive (disabled).

Parameters:

action (string, REQUIRED) Enum: [list, read, create, update, delete]  
name (string, optional) Skill name (required for all except list).  
content (string, optional) Skill text content (required for create).  
mode (string, optional) Enum: [disabled, enabled, on\_demand]

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None.

---

### 5.3 run\_instruction

---

Description:

Launch a saved agent instruction as a separate MyAgent process.  
Fire-and-forget. Use manage\_instructions(action='list') first.

Parameters:

name (string, REQUIRED) Name of the saved instruction.  
headless (boolean, optional) Run without GUI (default: true).

Implementation: subprocess.Popen with sys.executable to launch a new MyAgent.py process with -l "Name" [--headless].

PROVIDER DIFFERENCES: None.

OS DIFFERENCES:

|         |  |                                              |
|---------|--|----------------------------------------------|
| +-----+ |  | -----+                                       |
| Windows |  | subprocess.Popen with CREATE_NO_WINDOW flag. |
| +-----+ |  | -----+                                       |
| macOS   |  | subprocess.Popen without special flags.      |
| +-----+ |  | -----+                                       |

---

## 6. DYNAMIC TOOL: get\_skill

---

Description:

Retrieve the full content of an on-demand skill by name.

Parameters:

skill\_name (string, REQUIRED) Name of the skill to retrieve.  
Enum: (dynamically populated with all skills whose mode is "on\_demand")

Only added when at least one skill has mode="on\_demand". The enum values are generated dynamically from the current skills dictionary.

This is a PARALLEL\_SAFE tool.

PROVIDER DIFFERENCES: None.

OS DIFFERENCES: None.

---

## 7. SERVER-SIDE TOOLS BY PROVIDER

---

These tools are NOT defined in the local tool lists. They are added directly to each provider's API call and executed server-side by the provider's infrastructure. The model decides when to use them during generation.

---

### 7.1 Anthropic -- 2 Server-Side Tools

---

#### WEB SEARCH:

Name: web\_search  
Type: web\_search\_20250305  
Added as: {"type": "web\_search\_20250305", "name": "web\_search"}  
Beta flag: "web-search-2025-03-05"

The model searches the web directly during generation. Results appear as server\_tool\_use stream events. Provides richer, more current results than local DuckDuckGo search.

#### CODE EXECUTION:

Name: code\_execution  
Type: code\_execution\_20250825  
Added as: {"type": "code\_execution\_20250825", "name": "code\_execution"}  
Beta flags: "code-execution-2025-08-25", "files-api-2025-04-14"

The model writes and executes Python code in a sandboxed environment. Results (stdout + generated images) are extracted from final\_message.content AFTER streaming completes (file IDs are not available during streaming). Generated image files are downloaded via client.beta.files.download(file\_id), displayed inline in the chat (scaled to max 600px), and saved to saved\_chats/ as timestamped PNGs.

#### API call uses:

```
client.beta.messages.stream(
 betas=["web-search-2025-03-05",
 "code-execution-2025-08-25",
 "files-api-2025-04-14"],
 ...
)
```

---

### 7.2 OpenAI -- 2 Server-Side Tools

---

#### WEB SEARCH:

Type: web\_search\_preview  
Added as: {"type": "web\_search\_preview"}

The model searches the web during generation via the Responses API. Results are integrated into the streaming response.

#### CODE INTERPRETER:

Type: code\_interpreter  
Added as: {"type": "code\_interpreter", "container": {"type": "auto"}}

```
Include: ["code_interpreter_call.outputs"]
```

The model writes and executes code in a sandboxed container. Results stream as `code_interpreter_call` events. Image URLs are directly available in output objects. Generated images are displayed inline (scaled to max 600px) and saved to `saved_chats/` as timestamped PNGs. Files can also be downloaded via `openai_client.files.content(file_id)`.

API call uses:

```
openai_client.responses.create(
 tools=responses_tools,
 include=["code_interpreter_call.outputs"],
 store=False,
 ...
)
```

---

### 7.3 Gemini -- 0 Server-Side Tools

---

NONE.

Gemini CANNOT combine its built-in tools (`google_search`, `code_execution`) with custom function declarations. This is a fundamental API restriction.

Consequences:

- Web search uses the LOCAL DuckDuckGo implementation (5 results max).
- Code execution is NOT AVAILABLE for Gemini.
- All tools are local function calls processed by `MyAgent.py`.

This is the single largest capability difference between providers.

---

## 8. TOOL SCHEMA TRANSLATION BY PROVIDER

---

Each provider has different requirements for how tool schemas are formatted.

---

### 8.1 Anthropic -- Native Format (No Translation)

---

Tools use the Anthropic native format directly:

```
{
 "name": "tool_name",
 "description": "...",
 "input_schema": {
 "type": "object",
 "properties": { ... },
 "required": [...]
 }
}
```

No schema manipulation is needed.

---

## 8.2 OpenAI -- `_tools_to_responses()`

---

Converts to OpenAI Responses API function format:

```
{
 "type": "function",
 "name": tool["name"],
 "description": tool["description"],
 "parameters": tool["input_schema"], # renamed from input_schema
 "strict": False
}
```

Key differences from Anthropic:

- Wrapped in {"type": "function", ...} envelope.
- input\_schema renamed to parameters.
- strict: False flag added (non-strict schema validation).
- additionalProperties and integer enums are preserved (OpenAI handles them fine).

---

## 8.3 Gemini -- `_tools_to_gemini()`

---

Converts to `genai_types.FunctionDeclaration` objects with significant schema modifications:

1. STRIPS `additionalProperties`:  
Recursively removed from all schema levels. Gemini models reject schemas containing this field.
2. STRIPS non-string enums:  
Gemini only allows enum constraints on STRING type properties.
  - String enums: preserved (values cast to str).
  - Integer enums (e.g., `clicks: [1, 2]`): enum removed entirely, parameter reverts to plain integer type.
  - Rationale: "converting integer params to string type confuses Gemini's coordinate reasoning in desktop tools."
3. APPENDS coordinate hints (`_GEMINI_COORD_HINTS`):  
Desktop tool descriptions get additional text explaining the coordinate system. Appended to: `screenshot`, `mouse_click`, `mouse_scroll`, `mouse_drag`, `read_screen_text`. Example for `screenshot`:  
  
"CRITICAL COORDINATE RULES: The returned image has a specific pixel resolution (stated in the response text, e.g. 1920x1080). All coordinate-based tools use pixel positions within THIS image. Origin (0,0) is the TOP-LEFT corner. X increases rightward, Y increases downward..."
4. NORMALISES protobuf args (`_normalize_gemini_args`):  
Gemini's protobuf Struct returns all numbers as float64. Edge cases produce string-encoded numbers like "500.0" that break `int()` calls. After receiving tool call args, string numbers are converted back to int/float; real strings like "left" are left untouched.

Result format:

```
genai_types.FunctionDeclaration(
 name=tool["name"],
 description=desc + coord_hints,
 parameters=cleaned_schema
)
```

---

## 9. TOOL RESULT HANDLING BY PROVIDER

---

When a tool finishes executing, its result must be sent back to the model in the correct format for each provider.

---

### 9.1 Internal Format (Shared)

---

All tool results are initially stored in a common format:

```
{
 "type": "tool_result",
 "tool_use_id": block.id,
 "content": result_string_or_list
}
```

The `_ToolBlock` wrapper class normalises tool call objects so that OpenAI/Gemini dict-based tool blocks expose the same `.name`, `.id`, `.input` attribute interface as Anthropic's Pydantic objects. This allows `_execute_tool()` to work identically for all providers.

---

### 9.2 Anthropic -- Direct Format

---

Tool results are appended as a user message with the `tool_result` list:

```
messages.append({"role": "user", "content": [
 {"type": "tool_result", "tool_use_id": "...", "content": "..."},
 ...
]})
```

This is the native Anthropic format, used as-is.

---

### 9.3 OpenAI -- `_messages_to_responses()` Translation

---

Tool results are converted to `function_call_output` items:

```
{
 "type": "function_call_output",
 "call_id": tool_use_id,
 "output": text_content
}
```

Image content in tool results is preserved in the message parts. The `call_id` maps back to the original function call.

## 9.4 Gemini -- \_messages\_to\_gemini() Translation

Tool results are converted to `Part.from_function_response()` objects:

```
genai_types.Part.from_function_response(
 name=tool_name,
 response={"result": text_content}
)
```

Gemini tool call IDs are synthesised as "gemini\_{name}\_{index}" since the Gemini API does not use IDs natively. For thinking-capable models, `thought_signature` bytes from tool call parts are preserved and passed back in the function response to maintain the thinking chain.

## 10. RETRY & ERROR HANDLING BY PROVIDER

All providers retry up to 10 times. The retry behaviour differs:

| Error            | Anthropic                                                       | OpenAI                                                                                                     | Gemini                                                          |
|------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Rate limit (429) | Exponential backoff: $\min(2^a \cdot 5, 60s)$<br>~6.5 min total | Exponential backoff: $\min(2^a \cdot 5, 60s)$<br>~6.5 min total                                            | Exponential backoff: $\min(2^a \cdot 5, 60s)$<br>~6.5 min total |
| Server error     | 529 overload: $\min(2^a \cdot 10, 90s)$<br>~10 min total        | 500+ status: $\min(2^a \cdot 10, 90s)$<br>~10 min total                                                    | 500+ status: $\min(2^a \cdot 10, 90s)$<br>~10 min total         |
| Timeout          | N/A                                                             | <code>APITimeoutError</code> :<br>retry immediately<br>180s first-content timeout with 15s ticker messages | <code>TimeoutError</code> :<br>retry immediately                |
| Bad request      | N/A                                                             | <code>BadRequestError</code> (temperature):<br>removes temp param and retries                              | N/A                                                             |

All three providers break out of the retry loop immediately on success. The STOP button sets `stop_requested` which is checked inside the stream event loop for all providers, allowing mid-stream cancellation.

## =====

### 11. COMMAND SAFETY SYSTEM (OS-SPECIFIC)

## =====

The `run_command` tool checks every command against two pattern lists before execution: `COMMAND_BLOCKED` (rejected outright) and `COMMAND_CONFIRM` (requires user approval via a confirmation dialog).

Patterns use Python regex. The PS Safety dialog lets users disable specific `COMMAND_CONFIRM` patterns per instruction.

#### -----

#### 11.1 Windows -- `COMMAND_BLOCKED` (always rejected)

#### -----

`Format-Volume`, `Format-Disk`, `Clear-Disk`, `Initialize-Disk`, `Stop-Computer`, `Restart-Computer`, `Set-ExecutionPolicy`, `reg delete`, `Remove-ItemProperty` (`HKLM/HKCU`), `Remove-Item` (`HKLM/HKCU`), `bcdedit`, `diskpart`, `net user` (`/add|/delete`), `Disable-LocalUser`, `Remove-LocalUser`, `Clear-EventLog`, `wmic delete`

#### -----

#### 11.2 Windows -- `COMMAND_CONFIRM` (user approval required)

#### -----

`Remove-Item`, `del`, `rmdir`, `rm`, `rd`, `Clear-Content`, `Clear-RecycleBin`, `Stop-Process`, `kill`, `taskkill`, `Stop-Service`, `Remove-Service`, `Uninstall-Package`, `Move-Item`, `Rename-Item`, `Set-Content`, `Out-File`, `Invoke-Expression`, `iex`, `Invoke-WebRequest -OutFile`, `Start-Process`, `New-Service`, `-Recurse`, `-Force`

#### -----

#### 11.3 macOS -- `COMMAND_BLOCKED` (always rejected)

#### -----

`sudo rm -rf /` (bare root), `mkfs`, `dd of=/dev/*`, `shutdown`, `reboot`, `diskutil eraseDisk`, `diskutil partitionDisk`, `newfs`, `csrutil disable`, `dscl -delete`, `sysadminctl -deleteUser`

#### -----

#### 11.4 macOS -- `COMMAND_CONFIRM` (user approval required)

#### -----

`rm`, `mv`, `kill`, `killall`, `chmod`, `chown`, `sudo`, `curl -o`, `wget`, `launchctl`, `defaults write`, `defaults delete`, `brew install/uninstall/remove`, `pip install/uninstall`, `npm install/uninstall`, `open -a`, `diskutil`, `networksetup`, `pmset`

## =====

### 12. PLATFORM INITIALISATION & IMPORTS

## =====

Platform detection:



```
IS_WINDOWS = sys.platform == "win32"
```

Subprocess helper:

```
_SUBPROCESS_NOWND = {"creationflags": subprocess.CREATE_NO_WINDOW}
if IS_WINDOWS else {}
```

Windows-only initialisation:

- ctypes.windll.shcore.SetProcessDpiAwareness(2) for per-monitor DPI awareness (fallback: ctypes.windll.user32.SetProcessDPIAware()).
- import pygetwindow as gw (for window enumeration).
- ImageGrab from PIL (for multi-monitor screenshots with all\_screens).
- winocr (for Windows OCR).

macOS-only initialisation:

- mouseinfo pre-import hack: creates a stub module if mouseinfo import fails (prevents pyautogui import crash on some macOS setups).
- Quartz/CoreGraphics loaded on demand for:
  - CGWindowListCreateImage (screenshots)
  - CGWindowListCopyWindowInfo (window enumeration)
  - CGGetActiveDisplayList (display enumeration)
- Vision framework loaded on demand via objc.loadBundle() for OCR.
- osascript used for window activation via AppleScript.

Dialog behaviour:

- Windows: .transient(parent) is called on all dialogs for proper parent-child window relationship.
- macOS: .transient(parent) is SKIPPED on ALL dialogs. macOS's window manager restricts transient windows to the parent's screen, preventing them from being dragged to secondary monitors.

Multi-instance lock files:

- Windows: PID verification uses ctypes.windll.kernel32.OpenProcess with psapi.GetModuleBaseNameW to confirm the PID belongs to python.exe or pythonw.exe (not a recycled PID).
- macOS: PID verification uses os.kill(pid, 0) existence check.

Monitor config key (for geometry persistence):

- Windows: Win32 EnumDisplayMonitors to build a display config string.
- macOS: CoreGraphics CGGetActiveDisplayList for display config string.

### 13. PROVIDER COMPARISON TABLE

| Feature    | Anthropic   | OpenAI      | Gemini        |
|------------|-------------|-------------|---------------|
| Web Search | Server-side | Server-side | Local (DDGS)  |
|            | web_search_ | web_search_ | 5 results max |

|                           |                |               |               |
|---------------------------|----------------|---------------|---------------|
|                           | 20250305       | preview       |               |
| Code Execution            | Server-side    | Server-side   | NOT available |
|                           | code_exec_     | code_inter-   |               |
|                           | 20250825       | preter        |               |
| Code Exec Image Retrieval | Post-stream    | During stream | N/A           |
|                           | via file ID    | via URL/ID    |               |
|                           | beta files API |               |               |
| Local web_search kept?    | No             | No            | Yes           |
| Local fetch_webpage kept? | No             | No            | Yes           |
| Tool schema format        | Native         | Responses API | FunctionDecl  |
|                           | (no change)    | function wrap | objects       |
| additionalProperties      | Kept           | Kept          | Stripped      |
| Integer enums             | Supported      | Supported     | Stripped      |
| Coordinate hints in descs | No             | No            | Yes           |
| Protobuf number fixing    | Not needed     | Not needed    | Required      |
| Screenshot max size       | 1568px/1.15MP  | 2048px/2.0MP  | 2048px/2.0MP  |
| Beta flags required       | Yes (3)        | No            | No            |

|   |                           |             |               |               |
|---|---------------------------|-------------|---------------|---------------|
| + | strict schema mode        | N/A         | strict: false | N/A           |
| + | Tool call ID format       | Native UUID | Native ID     | Synthesised   |
|   |                           |             |               | gemini_name_N |
| + | Tool result format        | tool_result | function_call | Part.from_    |
|   |                           | (native)    | _output       | function_resp |
| + | _ToolBlock wrapper needed | No          | Yes           | Yes           |
| + | Thinking chain preserved  | Native      | reasoning     | thought_      |
|   |                           | thinking    | summary: auto | signature     |
|   |                           | blocks      |               | bytes         |
| + | First-content timeout     | None        | 180s + 15s    | None          |
|   |                           |             | ticker msgs   |               |
| + | Temperature auto-retry    | No          | Yes (removes  | No            |
|   |                           |             | temp on Bad-  |               |
|   |                           |             | RequestError) |               |
| + | Message format            | Native      | _messages_to_ | _messages_to_ |
|   |                           |             | responses()   | gemini()      |
| + |                           |             |               |               |

#### 14. OPERATING SYSTEM COMPARISON TABLE

| Feature                                                         | Windows                                              | macOS                               |
|-----------------------------------------------------------------|------------------------------------------------------|-------------------------------------|
| Platform check                                                  | sys.platform == "win32"                              | not IS_WINDOWS                      |
| DPI setup                                                       | SetProcessDpiAwareness(2)<br>(per-monitor DPI aware) | None needed                         |
| Shell for run_command                                           | PowerShell -NoProfile                                | /bin/bash -c                        |
| Subprocess window flag                                          | CREATE_NO_WINDOW                                     | None                                |
| Display enumeration<br>CGGetActiveDisplayList<br>ctypes)        | EnumDisplayMonitors<br>(ctypes.windll.user32)        | (CoreGraphics                       |
| Virtual screen bounds<br>rects                                  | GetSystemMetrics(76-79)                              | min/max of display                  |
| Full display screenshot<br>CGWindowList-<br>CreateImage(CGRect) | ImageGrab.grab(bbox,<br>all_screens=True)            | Quartz                              |
| Region screenshot<br>pyautogui.screenshot<br>Retina fix         | ImageGrab.grab(bbox,<br>all_screens=True)            | (region=...) +                      |
| DPI/Retina handling<br>logical,<br>if !=                        | DPI-aware: physical px<br>coords, no extra work      | Detect phys vs<br>resize to logical |
| OCR engine<br>VNRecognizeTextRequest                            | winocr (Windows OCR)<br>asyncio.run(recognize_pil)   | Vision framework                    |

|                                                  |                            |                    |
|--------------------------------------------------|----------------------------|--------------------|
| +-----+                                          |                            |                    |
| Window enumeration<br>CGWindowListCopyWindowInfo | pygetwindow                |                    |
| +-----+                                          |                            |                    |
| Window activation<br>AppleScript                 | pygetwindow .activate()    | osascript          |
| activate                                         | .restore() if minimised    | tell app to        |
| +-----+                                          |                            |                    |
| App launcher command<br>                         | "start <name>"             | "open -a '<Name>'" |
| +-----+                                          |                            |                    |
| Browser kill (browser_open)<br>"Microsoft Edge"  | taskkill /F /IM msedge.exe | pkill -f           |
| +-----+                                          |                            |                    |
| Browser paths<br>/Applications/...app/           | %ProgramFiles%\...\        |                    |
|                                                  | %ProgramFiles(x86)%\...\   | Contents/MacOS/... |
|                                                  | %LocalAppData%\...\        |                    |
| +-----+                                          |                            |                    |
| Paste shortcut (non-ASCII)<br>                   | Ctrl+V                     | Command+V          |
| +-----+                                          |                            |                    |
| Key aliases<br>>command                          | windows->win               | windows/win/cmd-   |
|                                                  | control->ctrl              | option->alt        |
|                                                  | return->enter              | control->ctrl      |
|                                                  | esc->escape                | return->enter,     |
| esc->escape                                      |                            |                    |
| +-----+                                          |                            |                    |
| Dialog transient()<br>restricts                  | Called (works across       | SKIPPED (macOS     |
| screen)                                          | screens on Windows)        | to parent's        |
| +-----+                                          |                            |                    |
| PID verification                                 | kernel32.OpenProcess +     | os.kill(pid, 0)    |
|                                                  | psapi.GetModuleBaseName    |                    |
| +-----+                                          |                            |                    |
| Blocked commands                                 | PowerShell-centric:        | Unix-centric:      |
|                                                  |                            |                    |

|                       |                             |                  |
|-----------------------|-----------------------------|------------------|
|                       | Format-Volume, bcdedit,     | sudo rm -rf /,   |
| mkfs,                 |                             |                  |
|                       | diskpart, net user, etc.    | dd of=/dev,      |
| shutdown, etc.        |                             |                  |
| +-----+               |                             |                  |
| Confirm commands      | PowerShell-centric:         | Unix-centric:    |
|                       | Remove-Item, Stop-Process,  | rm, mv, kill,    |
| chmod,                |                             |                  |
|                       | Set-Content, -Force, etc.   | sudo, brew, pip, |
| etc.                  |                             |                  |
| +-----+               |                             |                  |
| KNOWN_APPS extras     | notepad, notepad++,         | safari, terminal |
|                       | powerpoint, cmd, powershell |                  |
|                       |                             |                  |
| +-----+               |                             |                  |
| mouseinfo import hack | Not needed                  | Stub module      |
| created if            |                             | import fails     |
|                       |                             |                  |
| +-----+               |                             |                  |
| +-----+               |                             |                  |

## 15. TOOL COUNTS PER PROVIDER

With all toggles enabled and at least one on-demand skill:

ANTHROPIC (32 tools):

Local (30):

|                                                                      |      |
|----------------------------------------------------------------------|------|
| run_command, csv_search, user_prompt, get_skill                      | = 4  |
| + 13 DESKTOP_TOOLS                                                   | = 13 |
| + 11 BROWSER_TOOLS                                                   | = 11 |
| + 3 META_TOOLS (manage_instructions, manage_skills, run_instruction) | = 3  |

Subtotal: 31 local ... wait, 4+13+11+3 = 31 -- but web\_search and fetch\_webpage are EXCLUDED.

So: 31 - 0 = 31 local (the 5 base minus 2 excluded = 3, + 13+11+3+1)

Server-side (2):

web\_search\_20250305, code\_execution\_20250825

TOTAL: 33

OPENAI (33 tools):

Local (31): Same as Anthropic (web\_search/fetch\_webpage excluded).

Server-side (2): web\_search\_preview, code\_interpreter

TOTAL: 33

GEMINI (34 tools):

Local (34):

web\_search, fetch\_webpage, run\_command, csv\_search,

```
user_prompt, get_skill = 6
+ 13 DESKTOP_TOOLS = 13
+ 11 BROWSER_TOOLS = 11
+ 3 META_TOOLS = 3
Subtotal: 33 ... wait: 5 base + 1 get_skill + 13+11+3 = 33
Server-side (0): None.
TOTAL: 33
```

Note: Exact count is 33 for all providers when all toggles are on and one on-demand skill exists. The composition differs (2 server-side vs 2 extra local for Gemini), but the total is the same.

With minimal toggles (no Desktop, no Browser, no Meta, no on-demand skills):

```
Anthropic: 3 local + 2 server-side = 5
OpenAI: 3 local + 2 server-side = 5
Gemini: 5 local + 0 server-side = 5
```

```
=====
END OF REFERENCE
=====
```